

Youngsoo Shin

professor

基本信息

姓名: Youngsoo Shin

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学科领域

电子与计算机工程

集成电路设计与集成系统

微电子科学与工程

研究领域

集成电路设计与制造

数字集成电路设计

EDA工具与电路验证

电路与系统设计

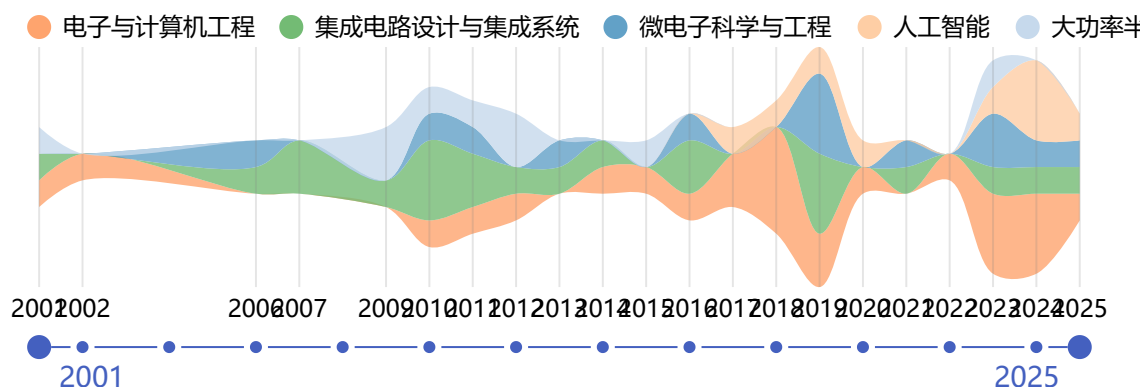
低功耗电路与能量管理

超大规模集成电路设计

个人简介

Youngsoo Shin is a Professor of Electrical Engineering at KAIST, with a verified email at ee.kaist.ac.kr. His research interests include CADEDAVLSI Low power. He has an overall citation count of 4041 and an h - index of 27. Since 2020, his citations are 902 and h - index is 16. He has published numerous influential articles, such as "Power conscious fixed priority scheduling for hard real - time systems" in 1999, which has been cited 512 times.

近年发表的论文



Model-Based OPC With Adaptive PID Control Through Reinforcement Learning

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Kim, Taeyoung; Zhang, Shilong; Shin, Youngsoo

被引用次数: 0

2025

ML-Guided Curvilinear OPC: Fast, Accurate, and Manufacturable Curve Correction

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Kim, Seohyun; Zhang, Shilong; Shin, Youngsoo

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2025

Integrated Netlist Synthesis and In-Memory Mapping for Memristor-Aided Logic

PROCEEDING OF THE GREAT LAKES SYMPOSIUM ON VLSI 2024, GLSVLSI 2024

Lee, Seunggyu; Lee, Wonjae; Shin, Youngsoo

被引用次数: 0

2024

A Fast and Accurate PEB Simulation Through Recurrent Neural Network

DTCO AND COMPUTATIONAL PATTERNING III

Cho, Gangmin; Kim, Taeyoung; Kim, Seohyun; Shin, Youngsoo

被引用次数: 0

2024

Fast IR-Drop Prediction of Analog Circuits Using Recurrent Synchronized GCN and Y-Net Model

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Lee, Seunggyu; Hyun, Daijoon; Jung, Younggwang; Cho, Gangmin; Shin, Youngsoo

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2024

Accurate Interpolation of Library Timing Parameters Through Recurrent Convolutional Neural Network

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Hyun, Daijoon; Jung, Younggwang; Shin, Youngsoo

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2024

Decap Insertion With Local Cell Relocation Minimizing IR-Drop Violations and Routing DRVs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hyun, Daijoon; Jung, Younggwang; Shin, Youngsoo

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2024

Lightning Talk 21: EDA with ML, Rule-Based, or Both?

Design Automation Conference

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2023

Power Distribution Network Optimization Using HLA-GCN for Routability Enhancement

2023 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Jung, Younggwang; Hyun, Daijoon; Choi, Soyeon; Shin, Youngsoo

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2023

Airgap Insertion and Layer Reassignment Under Setup and Hold Timing Constraints

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

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2023

Integrated Power Distribution Network Synthesis for Mixed Macro Blocks and Standard Cells

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Hyun, Daijoon; Lee, Wonjae; Park, Jinhyeong; Shin, Youngsoo

被引用次数: 2

2023

Routability Optimization of Extreme Aspect Ratio Design through Non-uniform Placement Utilization and Selective Flip-flop Stacking

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Hyun, Daijoon; Koh, Sunwha; Jung, Younggwang; Kim, Taeyoung; Shin, Youngsoo

被引用次数: 0

2023

Fast Optical Proximity Correction Using Graph Convolutional Network With Autoencoders

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Cho, Gangmin; Kim, Taeyoung; Shin, Youngsoo

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2023

Calibration of Compact Resist Model Through CNN Training

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Kwon, Yonghwi; Shin, Youngsoo

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2023

Integrated Test Pattern Extraction and Generation for Accurate Lithography Modeling

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Cho, Gangmin; Kwon, Yonghwi; Kareem, Pervaiz; Shin, Youngsoo

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2022

Synthesis of Lithography Test Patterns Using Machine Learning Model

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Kareem, Pervaiz; Shin, Youngsoo

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2021

Optical Proximity Correction Using Bidirectional Recurrent Neural Network With Attention Mechanism

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Kwon, Yonghwi; Shin, Youngsoo

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2021

Layout Pattern Synthesis for Lithography Optimizations

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Kareem, Pervaiz; Kwon, Yonghwi; Shin, Youngsoo

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2020

Cut Optimization for Redundant Via Insertion in Self-Aligned Double Patterning

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Song, Youngsoo; Hyun, Daijoon; Lee, Jingon; Jung, Jinwook; Shin, Youngsoo

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2019

Integrated Approach of Airgap Insertion for Circuit Timing Optimization

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Hyun, Daijoon; Shin, Youngsoo

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2019

Integrated Latch Placement and Cloning for Timing Optimization

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Jung, Jinwook; Nam, Gi-Joon; Chung, Woohyun; Shin, Youngsoo

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2019

Neural Network Classifier-Based OPC With Imbalanced Training Data

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Choi, Suhyeong; Shim, Seongbo; Shin, Youngsoo

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2019

Selective Use of Stitch-Induced Via for V0 Mask Reduction: Standard Cell Design and Placement Optimization

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Hyun, Daijoon; Jung, Younggwang; Shin, Youngsoo

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2019

Memory-Efficient Parametric Semiglobal Matching

IEEE SIGNAL PROCESSING LETTERS

Lee, Yeongmin; Park, Min-Gyu; Hwang, Youngbae; Shin, Youngsoo; Kyung, Chong-Min

被引用次数: 19

2018

Module grouping to reduce the area of test wrappers in SoCs

INTEGRATION-THE VLSI JOURNAL

Kim, Sangmin; Shin, Youngsoo

被引用次数: 0

2018

Recap of the 23rd Asia and South Pacific Design Automation Conference (ASP-DAC)

IEEE DESIGN & TEST

Shin, Youngsoo

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2018

Electrothermal Analysis With Nonconvective Boundary Conditions

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Suhyeong; Shim, Seongbo; Shin, Youngsoo

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2018

OWARU: Free Space-Aware Timing-Driven Incremental Placement With Critical Path Smoothing

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Jung, Jinwook; Nam, Gi-Joon; Reddy, Lakshmi N.; Jiang, Iris Hui-Ru; Shin, Youngsoo

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2018

Machine Learning-Guided Etch Proximity Correction

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

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2017

Lithography Defect Probability and Its Application to Physical Design Optimization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Shim, Seongbo; Chung, Woohyun; Shin, Youngsoo

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2017

Fast Verification of Guide-Patterns for Directed Self-Assembly Lithography

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2017

Reliable Memristive Switching Memory Devices Enabled by Densely Packed Silver Nanocone Arrays as Electric -Field Concentrators

ACS NANO

You, Byoung Kuk; Kim, Jong Min; Joe, Daniel J.; Yang, Kyoungsoon; Shin, Youngsoo; Jung, Yeon Sik; Lee, Keon Jae

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Light Interference Map: A Prescriptive Optimization of Lithography-Friendly Layout

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2016

Wakeup scheduling and its buffered tree synthesis for power gating circuits

INTEGRATION-THE VLSI JOURNAL

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2016

One-Cycle Correction of Timing Errors in Pipelines With Standard Clocked Elements

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Shin, Insup; Kim, Jae-Joon; Lin, Yu-Shiang; Shin, Youngsoo

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2016

Synthesis of Dual-Mode Circuits Through Library Design, Gate Sizing, and Clock-Tree Optimization

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Kim, Sangmin; Kang, Seokhyeong; Shin, Youngsoo

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2016

Topology-oriented pattern extraction and classification for synthesizing lithography test patterns

JOURNAL OF MICRO-NANOLITHOGRAPHY MEMS AND MOEMS

Shim, Seongbo; Shin, Youngsoo

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2015

An Analytical Approach to Thermal Design and Optimization With a Temperature-Dependent Power Model

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shim, Seongbo; Lee, Jae Wook; Shin, Youngsoo

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Aggressive Voltage Scaling Through Fast Correction of Multiple Errors With Seamless Pipeline Operation

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HAPL: Heterogeneous Array of Programmable Logic Using Selective Mask Patterning

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Shin, Youngsoo; Shin, Insup; Baek, Donkyu; Kim, Duckhwan; Paik, Seungwhun

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Simplifying Clock Gating Logic by Matching Factored Forms

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Han, Inhak; Shin, Youngsoo

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Maximizing Frequency and Yield of Power-Constrained Designs Using Programmable Power-Gating

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SAMPLING CORRELATION SOURCES FOR TIMING YIELD ANALYSIS OF SEQUENTIAL CIRCUITS WITH CLOCK NETWORKS

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Yu, Lee-Eun; Shin, Changsik; Paik, Seungwhun; Liou, Jing-Jia; Shin, Youngsoo

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Pulse Width Allocation and Clock Skew Scheduling: Optimizing Sequential Circuits Based on Pulsed Latches

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Lee, Hyein; Paik, Seungwhun; Shin, Youngsoo

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LOOKUP TABLE-BASED ADAPTIVE BODY BIASING OF MULTIPLE MACROS FOR PROCESS VARIATION COMPENSATION AND LOW LEAKAGE

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Supply Switching With Ground Collapse for Low-Leakage Register Files in 65-nm CMOS

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2010

Power Gating: Circuits, Design Methodologies, and Best Practice for Standard-Cell VLSI Designs

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Shin, Youngsoo; Seomun, Jun; Choi, Kyu-Myung; Sakurai, Takayasu

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2010

Minimizing Leakage Power of Sequential Circuits through Mixed-Vt Flip-Flops and Multi-Vt Combinational Gates

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Semicustom Design of Zigzag Power-Gated Circuits in Standard Cell Elements

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Supply switching with ground collapse: Simultaneous control of subthreshold and gate leakage current in nanometer-scale CMOS circuits

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Semicustom design methodology of power gated circuits for low leakage applications

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2007

Power analysis of VLSI interconnect with RLC tree models and model reduction

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Shin, Youngsoo; Lee, Junghyup

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Exercise improves bone mineral density during treatment of postmenopausal osteoporosis with alendronate.

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Early analysis tools for system-on-a-chip design

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Partial bus-invert coding for power optimization of application-specific systems

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Narrow bus encoding for low-power DSP systems

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Airgap Insertion and Layer Reassignment Under Setup and Hold Timing Constraints

TCAD

Daijoon Hyun; Younggwang Jung; Youngsoo Shin

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Accurate Interpolation of Library Timing Parameters Through Recurrent Convolutional Neural Network

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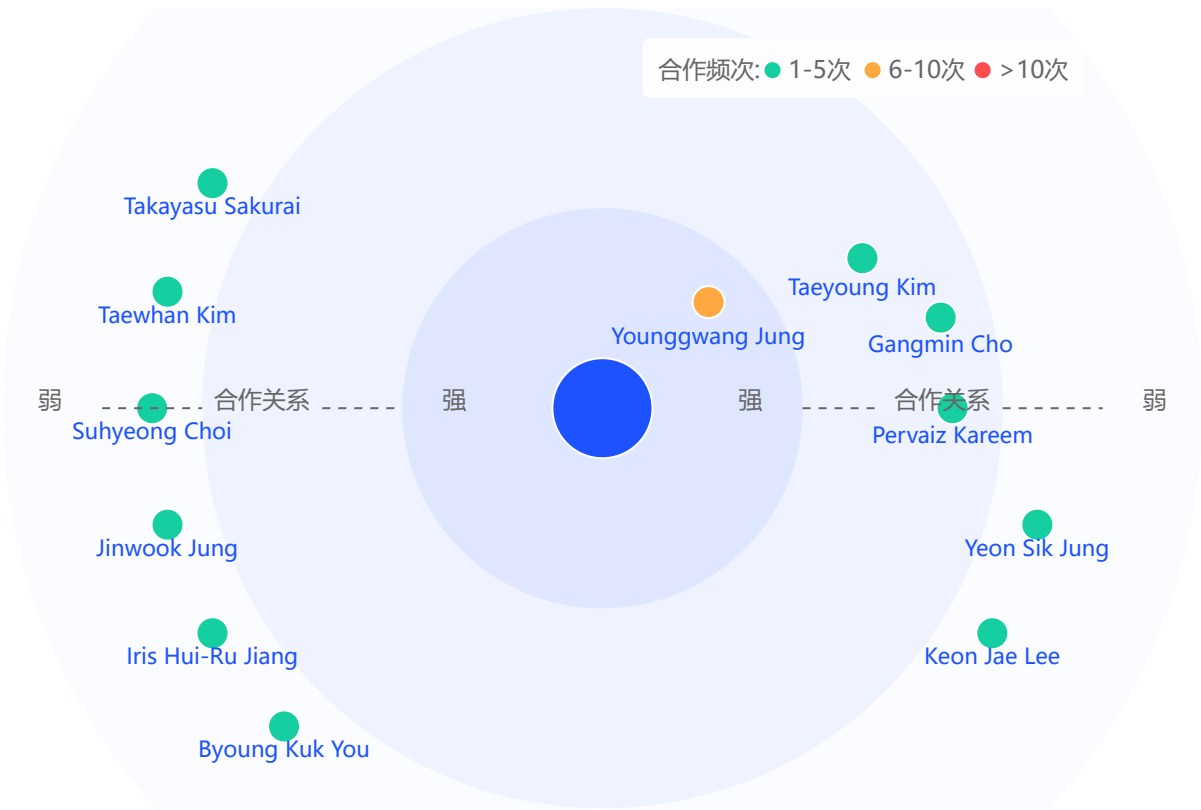
Daijoon Hyun; Younggwang Jung; Youngsoo Shin
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Fast Verification of Guide-Patterns for Directed Self-Assembly Lithography
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Seongbo Shim; Youngsoo Shin
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OWARU: Free Space-Aware Timing-Driven Incremental Placement With Critical Path Smoothing
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Jinwook Jung; Gi-Joon Nam; Lakshmi N. Reddy; Iris Hui-Ru Jiang; Youngsoo Shin
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Neural Network Classifier-Based OPC With Imbalanced Training Data
TCAD
Suhyeong Choi; Seongbo Shim; Youngsoo Shin
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学术合作



Younggwang Jung 合作次数: 7
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Taeyoung Kim 合作次数: 4

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Gangmin Cho

合作次数: 3

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